

AI Reddit Trend & Post Generator — Product Requirements Document

1. Executive Summary

The AI Reddit Trend & Post Generator is an AI-powered agent designed for content creators and marketers. It automates the discovery of trending topics on Reddit and generates relevant, engaging post drafts, significantly boosting content relevance and creation efficiency.

2. Problem Statement

- Content creators and marketers struggle to keep pace with rapidly changing trends on Reddit.
- Manually sifting through Reddit for trending topics is time-consuming and inefficient.
- Missed opportunities for timely and relevant content creation lead to reduced engagement.

3. Goals & Success Metrics

Goals:

- Increase the timeliness and relevance of content created by users.
- Significantly reduce the manual effort and time spent on trend research.
- Enhance user satisfaction with the quality and utility of AI-generated content.

Success Metrics:

- Trend Discovery: Increase in the number of trending topics identified per day by 20%.
- Efficiency: Reduction in time spent by users on trend research by 30%.
- Quality: Average user satisfaction score of 4/5 for generated post quality.

4. Target Users

Social Media Manager: Manages multiple social platforms, constantly needing fresh, engaging content. They struggle to keep up with Reddit's fast-paced trends, leading to missed opportunities for timely posts that could drive engagement.

Content Creator: Seeks new ideas and wants to leverage trending topics to connect

with their audience. They find manual trend research tedious and time-consuming, which can hinder their creative output and ability to publish consistently.

5. Scope

In Scope

- Automated discovery of trending posts from Reddit.
- AI-powered extraction of core trending topics and keywords.
- Generation of AI-drafted Reddit-style post titles and bodies.
- A user-friendly web interface to view trends and manage generated post drafts.

Out of Scope

- Direct publishing of posts to Reddit or any other social media platform.
- Advanced analytics on the performance of published posts.
- Integration with multiple social media platforms beyond Reddit.
- User-specific customization of AI generation parameters (e.g., specific persona, writing style, length control).

6. Core Features (MVP)

- Reddit Trending Post Discovery — Automatically identifies trending posts.
- Connects to the Reddit API to fetch posts from specified subreddits.
- Identifies and lists posts as 'trending' based on a defined algorithm (e.g., top 100 posts by upvotes in 24 hours).
- Trending posts include title, URL, subreddit, upvotes, and comment count.
- Trending Topic Extraction — Uses AI to distill overarching themes.
- Processes trending posts to identify distinct topics with concise names and relevant keywords.
- Associates multiple trending posts with a single extracted topic.
- Topic extraction accuracy is at least 80% as judged by human review.
- AI Post Generation (Draft) — Generates relevant and engaging post drafts.
- Given a trending topic, the system generates a unique post title and body.
- Generated posts are grammatically correct, coherent, and relevant to the selected topic.
- Users can view the generated post draft within the application.
- User Interface for Trend & Post Management — Provides a central hub for users.
- Displays a list of currently trending topics.

- Users can select a trending topic and initiate post generation.
- The UI displays generated post drafts for review, allowing users to copy content.

Post-MVP:

- Configurable Subreddit Monitoring
- Post Editing & Refinement

7. Key User Flows

1. Discover Trending Topics
- Content Creator logs into the application.
 - System automatically displays a dashboard with the latest trending topics.
 - Content Creator reviews the list of trending topics, including associated keywords.
2. Generate a New Post Draft
- Content Creator views the list of trending topics.
 - Content Creator selects a specific trending topic of interest.
 - Content Creator clicks 'Generate Post' button.
 - System displays the AI-generated post draft (title and body) in the UI.
 - Content Creator reviews the generated post.

8. Non-Functional Requirements

Category	Requirement	Target
Performance	Trending topic discovery should complete within 15 minutes.	
	Trend discovery latency < 15 minutes	
Performance	AI post generation should complete within 10 seconds per request.	
	Post generation latency < 10 seconds	
Security	All user authentication and API keys must be stored and transmitted securely.	
	Adherence to OWASP Top 10	
Scalability	The system should support up to 10,000 daily active users.	
	10,000 Daily Active Users (DAU)	

9. Open Questions & Risks

#	Question / Risk	Status
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| 1 | What is the precise algorithm or criteria for defining a 'trending' post? | Open
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| 2 | What is the desired tone and style for the AI-generated posts? | Open |

| 3 | Reliance on Reddit API terms of service, rate limits, and data availability. | Open
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| 4 | Cost implications of using external Large Language Model (LLM) APIs. |
Open |

| 5 | Accuracy and quality of AI-generated content are subject to LLM capabilities. |
Open |